

Essentials of Pediatrics: Ghai Principles

Pediatrics — MBBS 4th and Final Year

Standard Core Curriculum for Medical Undergraduates • Comprehensive Clinical Reference

Curriculum Year	MBBS 4th and Final Year
Medical Subject	Pediatrics
Target Audience	MBBS Undergraduates, Postgraduates, USMLE/PLAB Aspirants
Format	Fully Indexed Textbook with Chapter, Topic, and Page Referencing
Accreditation	Standard Medical Council Curriculum Guidelines

Chapter 1: Neonatal Care and Pediatric Infections

Neonatal Sepsis and Pediatric Bacterial Meningitis: Pathogens, Bulging Fontanelle, and Regimens

Pediatric bacterial meningitis carries high morbidity and mortality. In neonates (< 1 month), pathogens are acquired from the maternal genital tract: Group B Streptococcus (S. agalactiae, 50%), Escherichia coli (K1 strain), and Listeria monocytogenes. In infants and children (> 2 months), Streptococcus pneumoniae and Neisseria meningitidis predominate.

Clinical presentation in infants is often subtle and non-specific: hypothermia or fever, poor feeding, lethargy, high-pitched cry, vomiting, irritability, seizures, and a tense Bulging Anterior Fontanelle. Classic signs like neck rigidity, Kernig's, and Brudzinski's signs are typically ABSENT in infants under 12-18 months due to open fontanelles and immature motor responses.

Immediate lumbar puncture and blood cultures are required. Empiric neonatal regimen is IV Ampicillin (to cover Listeria and GBS) PLUS IV Cefotaxime or Gentamicin. In older infants (> 2 months), IV Ceftriaxone (100 mg/kg/day) PLUS IV Vancomycin (60 mg/kg/day) is administered. Dexamethasone is beneficial in H. influenzae type b and pneumococcal meningitis if given with the first dose.

Clinical Pearl: Classic signs of meningeal irritation (Kernig/Brudzinski) are absent in infants <12-18 months; a tense bulging anterior fontanelle, lethargy, and poor feeding are the primary clinical indicators of meningitis. Empiric neonatal therapy requires Ampicillin + Cefotaxime/Gentamicin.

	Clinical Hallmark Signs	Empiric Antibiotics
Signs and Symptoms	Hypothermia, lethargy, bulging anterior fontanelle, poor feeding	Ampicillin (150-200 mg/kg/day)
	Fever, vomiting, irritability, bulging fontanelle, seizures	Ceftriaxone (100 mg/kg/day)
	Fever, severe headache, photophobia, nuchal rigidity, Kernig/Brudzinski	Ceftriaxone + Vancomycin

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Pediatric Acute Diarrhea, Dehydration Assessment and WHO Oral Rehydration Protocols

Acute diarrhea is passage of ≥ 3 loose stools per day lasting < 14 days, primarily caused by Rotavirus, enterotoxigenic E. coli (ETEC), and Shigella. The leading cause of death is dehydration and electrolyte loss.

WHO classifies dehydration into three stages: (1) No Dehydration ($< 5\%$ fluid deficit): alert, normal eyes, drinks normally, skin pinch retracts immediately; (2) Some Dehydration (5-10% deficit): restless, irritable, sunken eyes, drinks eagerly/thirsty, skin pinch goes back slowly ($< 2s$); (3) Severe Dehydration ($> 10\%$ deficit): lethargic or unconscious, very sunken eyes, unable to drink, skin pinch goes back very slowly ($> 2s$).

Management Protocols: Plan A (No dehydration): home fluids, zinc supplementation (20 mg/day for 14 days; 10 mg for < 6 months), continue feeding. Plan B (Some dehydration): Low-osmolarity Oral Rehydration Salt (ORS) solution (75 mL/kg over 4 hours) plus zinc. Plan C (Severe dehydration): emergency IV fluids with Ringer's Lactate (100 mL/kg divided into rapid initial and slower maintenance infusions according to age: < 12 months vs ≥ 12 months).

Clinical Pearl: WHO Low-osmolarity ORS has total osmolarity of 245 mOsm/L (Na⁺ 75, Glucose 75, K⁺ 20, Cl⁻ 65, Citrate 10). Zinc supplementation for 14 days reduces duration and recurrence.

	Estimated Fluid Deficit	Treatment Protocol
te	$< 5\%$ (< 50 mL/kg)	Plan A: Home fluids
h slow ($< 2s$)	5% to 10% (50 - 100 mL/kg)	Plan B: ORS 75 mL/kg over 4 hours
pinch $> 2s$	$> 10\%$ (> 100 mL/kg)	Plan C: IV Ringer's Lactate